

EXHIBIT E-3.34(a)

TRUE-SOURCE DEPENDENCY ASSESSMENT (TSDA)

Methodology and Data Specification for Evaluating U.S. Critical-Mineral Supplier Diversification

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Initial Case Studies	Gallium and Germanium
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1. Research Objective

The True-Source Dependency Assessment (TSDA) is a quantitative method for determining whether diversification in the countries directly supplying a critical mineral to the United States corresponds to a reduction in U.S. dependence on concentrated foreign production and processing sources.

The initial application examines gallium and germanium before and after the Chinese export restrictions that began in 2023.

The principal research question is:

Did diversification in U.S. gallium and germanium suppliers reduce underlying foreign dependence across production and processing stages, or primarily change the countries recorded as direct suppliers?

2. Analytical Framework

TSDA evaluates dependence across four connected analytical components:

- Direct U.S. supplier
- Production source
- Processing source
- Intermediary tracing between the direct supplier and upstream sources

This structure permits TSDA to distinguish the country recorded as the U.S. supplier from the countries that actually supplied, produced, or processed the material upstream.

2.1 Direct U.S. Supplier

This stage identifies the countries recorded as supplying the United States. For each mineral and year, the dataset will include:

- supplier country;
- import quantity;
- customs value;
- supplier share of total U.S. imports;
- number of supplier countries;
- dominant supplier share; and
- supplier concentration.

The primary source will be USITC DataWeb, using the most detailed applicable U.S. Harmonized Tariff Schedule classification. [5]

For gallium metal, the current U.S. classification is HTS 8112.92.1000 - Gallium metal. USGS identifies the same tariff line in its 2026 gallium summary. [2]

For germanium, relevant current classifications include:

- 2825.60.0010 - Germanium dioxide
- 8112.92.6000 - Germanium metal, unwrought
- 8112.92.6500 - Germanium metal, powder
- 8112.99.1000 - Germanium metal, wrought

Germanium product categories will initially be analyzed separately and combined only where a documented conversion to a common germanium-content basis is available. [3,4]

Because tariff classifications can change over time, the applicable HTS line will be verified separately for each year from 2019 through 2025. Any change in classification will be documented in the data-source inventory together with the concordance used to construct a consistent time series.

Scope of Gallium Products

The principal TSDA dependency analysis will focus initially on gallium metal rather than gallium arsenide (GaAs) wafers. USGS separately tracks doped and undoped GaAs wafers under distinct tariff classifications. [2]

GaAs wafers are downstream semiconductor materials rather than the primary gallium supply form being traced in the initial true-source analysis. They will therefore be excluded from the core supplier-concentration calculation but may be examined as a secondary downstream response indicator during empirical validation.

2.2 Production Source

This stage identifies the countries in which the mineral is produced. Primary sources will include:

- U.S. Geological Survey Mineral Commodity Summaries;
- U.S. Geological Survey Minerals Yearbook; and
- associated USGS production datasets.

Country production shares and production concentration will be calculated where reliable country-level estimates are available.

For gallium, USGS reports that China accounted for 99% of worldwide primary low-purity gallium production in 2025. Japan and Russia were the remaining reported primary producers outside China. [2]

For germanium, reliable country-level production information is less complete. USGS reports 2025 U.S. net import reliance as greater than 50%, rather than as a precise percentage. [3]

Accordingly, germanium net import reliance will be recorded as the published bound ">50%," and production stages lacking sufficiently reliable country-level estimates will be classified as unresolved.

2.3 Processing Source

This stage identifies where relevant refining or processing occurs before the material reaches downstream U.S. users. Sources will include USGS mineral reports, government geological publications, documented refining or processing-capacity information, and other authoritative mineral-market sources.

Where reliable country-level processing shares are available, processing concentration will be calculated. Where the evidence establishes processing location but does not support a defensible quantitative share, the result will be reported qualitatively and identified as a data limitation.

2.4 Intermediary Tracing

Direct U.S. import statistics do not necessarily identify the original source of a mineral. A country may export material to the United States that it previously imported from another country.

Brainard, Gulley, and Nassar (2026) address this problem by constructing mineral trade networks that distinguish producing countries from intermediary or transshipment countries and trace direct and indirect trade pathways back to original producers. [1] TSDA will apply this provenance principle to the gallium and germanium cases.

For the principal analysis, intermediary tracing will be conducted for:

- each country accounting for at least 10% of U.S. imports of the relevant material in either comparison period; and
- any additional supplier required to account for at least 90% of total identified U.S. import quantity.

A 5% supplier threshold will be used as a sensitivity test.

For each selected U.S. supplier, TSDA will identify that country's own sources of the relevant material using the most detailed official national trade statistics available. Primary sources will include:

- Statistics Canada's Canadian International Merchandise Trade application, which provides imports at the 10-digit Canadian level; [6]

- Japan Customs Trade Statistics, which uses 9-digit national statistical codes built from the internationally harmonized 6-digit HS classification; [7]
- Eurostat Comext, which provides detailed EU merchandise-trade data by reporting country, partner, value, quantity, and detailed product classification; [8]
- UN Comtrade as a secondary source where sufficiently detailed national data are unavailable.

Domestic production in an intermediary country will be treated as a supply input alongside that country's imports. Where possible, the relative contributions of domestic production and imported material will be used to attribute the country's onward supply.

For each intermediary, upstream origin shares will be calculated from identifiable supply inputs. U.S. imports received through that country will then be proportionally attributed to those upstream sources.

For example, if 70% of the traceable gallium supply available to Supplier Country B originates from Country A and 30% originates from domestic or other sources, the corresponding U.S. flow through Country B will be attributed using those proportions rather than automatically treating Country B as an independent source.

These attributed flows will produce a true-source supplier distribution and a corresponding true-source HHI.

Limitation of Six-Digit Trade Data

Where detailed national statistics are unavailable, international trade data may be available only at the 6-digit Harmonized System level.

The United Nations Statistics Division confirms that HS 8112.92 combines gallium and germanium with hafnium, indium, niobium, rhenium, and vanadium, including unwrought forms, waste and scrap, and powders. [9]

Accordingly, an HS 8112.92 bilateral flow that cannot be separated into the target mineral will not be treated as a precise gallium or germanium quantity. It may be used as an approximate or upper-bound indicator and will be identified accordingly.

3. Study Period

The initial analysis will compare:

Pre-shock baseline: 2019-2022

Post-shock period: 2023-2025

The baseline establishes U.S. sourcing and upstream dependency immediately before China's gallium and germanium export controls. The post-shock period captures the controls introduced in 2023, subsequent sourcing adjustments, the December 2024 restrictions affecting exports to the United States, and the first complete calendar year following those restrictions. [2,3]

Period Aggregation

The principal baseline measure will use pooled quantities for 2019-2022:

$$\text{Baseline supplier share}(i) = \text{Sum}[2019-2022 \text{ imports}(i)] / \text{Sum}[2019-2022 \text{ total imports}]$$

The post-shock measure will similarly use pooled quantities for 2023-2025. HHI will be calculated from these pooled-period shares.

Annual supplier shares and annual HHIs will also be retained and reported as a time series to determine whether the pooled-period result conceals significant year-to-year changes. The same pooled-period approach will be applied to production and processing quantities where comparable annual data are available.

4. Core Measures

4.1 Country Supply Share

$$\text{Supplier Share}(i) = \text{Imports}(i) / \text{Total U.S. Imports}$$

Physical quantity will be the principal measure where units are comparable. Customs value will be retained for supplementary analysis and calculation of unit values.

4.2 Herfindahl-Hirschman Index

$$HHI = \text{Sum of squared country shares}$$

Country shares will be expressed in percentage points. On this scale, HHI approaches 0 as supply becomes increasingly dispersed and reaches 10,000 when a single country accounts for the entire supply.

For descriptive context only, the U.S. Department of Justice and Federal Trade Commission currently describe an HHI above 1,800 as highly concentrated in their merger-analysis framework. [10] TSDA does not use an antitrust threshold to determine whether mineral diversification is genuine.

HHI will be calculated, where data permit, for direct U.S. suppliers, true-source-adjusted suppliers, mineral production, and processing or refining.

4.3 Dominant-Country Share

TSDA will report the share supplied or controlled by the largest country at each stage. This provides a direct measure of whether reliance on the dominant source changed even where the overall distribution of suppliers also changed.

4.4 True-Source Concentration

$$\text{True-Source HHI} = \text{Sum of squared attributed upstream-source shares}$$

The difference between direct-supplier HHI and true-source HHI measures the extent to which apparent supplier diversity changes after intermediary relationships are taken into account.

4.5 Net Import Reliance

Official USGS net-import-reliance estimates will be used where available. Net import reliance and geographic concentration will remain separate measures:

- Net import reliance measures how dependent U.S. consumption is on imports.
- HHI measures how geographically concentrated the relevant supply is.

USGS reports gallium as 100% net import reliant in 2025. [2] Germanium is reported as greater than 50% net import reliant; TSDA will preserve that bound rather than convert it into an unsupported point estimate. [3]

5. Measuring Change

$$\text{Delta HHI} = \text{HHI}(\text{post}) - \text{HHI}(\text{baseline})$$

$$\text{Delta DS} = \text{DS}(\text{post}) - \text{DS}(\text{baseline})$$

DS represents the dominant-country share. A negative value indicates declining concentration and a positive value indicates increasing concentration.

Robustness Requirement

A supply stage will be treated as showing a decline in concentration only where:

1. Delta HHI is negative under the principal specification; and
2. the direction of the change remains negative under the specified sensitivity analyses.

If the sign of the change reverses under reasonable alternative assumptions, the evidence for a decline will be reported as sensitive to specification rather than classified as a robust reduction.

Sensitivity analysis will include, where applicable:

- a 5% rather than 10% intermediary-screening threshold;
- alternative treatment of uncertain upstream-source shares;
- annual HHI patterns alongside pooled-period HHI; and
- alternative documented conversion assumptions for product categories requiring conversion to mineral content.

6. TSDA Classification Framework

6.1 Genuine Reduction in Underlying Dependence

Direct U.S. supplier concentration declines and the true-source, production, and relevant processing evidence also shows a robust decline in upstream concentration.

6.2 Partial Reduction in Underlying Dependence

Direct sourcing becomes less concentrated and underlying dependence declines at one relevant upstream stage but not at another.

6.3 Apparent Diversification

Direct U.S. supplier concentration declines, but intermediary tracing or upstream production or processing evidence shows that dependence remains concentrated in substantially the same underlying source.

6.4 No Diversification

Direct U.S. supplier concentration does not decline under the principal and sensitivity specifications.

6.5 Unresolved

Available evidence is insufficient to determine whether dependence changed at one or more supply-chain stages required for classification.

7. Data Treatment and Reproducibility

- Raw source files will be preserved separately from transformed datasets.
- Annual comparisons will use complete calendar years.
- Physical quantity will be preferred for supply-flow measurement where reporting units are comparable.
- Customs value and unit value will be retained as supplementary measures.
- Product categories will remain separate unless a documented conversion permits aggregation.
- HTS and national tariff classifications will be verified for each study year, and any code changes or concordances will be recorded.
- Conflicting values reported by authoritative sources will be documented.
- Production or processing stages without sufficiently reliable evidence will be reported as unresolved.
- Every transformation from original source data to final output will be recorded in the data dictionary or processing log.

8. Empirical Validation

The 2023 and 2024 gallium and germanium restrictions provide the initial empirical test of TSDA. Validation will proceed in four steps:

1. Measure pre-shock dependence using the 2019-2022 direct-supplier, intermediary, production, and processing data.
2. Generate pre-specified dependency classifications before evaluating the complete post-shock outcome.
3. Measure the post-shock response using 2023-2025 import quantity, supplier participation, concentration, and unit values.
4. Compare the observed response with the prediction implied by the pre-shock classification.

8.1 Testable Predictions

If a direct supplier is classified as an intermediary materially dependent on the restricted upstream source, the post-restriction U.S. trade route is expected to exhibit at least one of the following relative to its pre-shock baseline:

- declining import quantity;
- rising unit value; or
- exit from the U.S. supplier set.

If a supplier is classified as independent of the restricted upstream source, its U.S. supply is expected to remain comparatively stable or expand as sourcing adjusts.

A classification will be considered supported where the observed post-shock pattern is consistent with its pre-specified prediction.

A classification will be considered contradicted where routes classified as dependent intermediaries remain stable without the predicted disruption while routes classified as independent exhibit the contraction.

Contradicted classifications will be reported, and the relevant intermediary-attribution rule, dependency measure, or classification rule will be reevaluated.

9. Primary Data Architecture

Variable	Measure	Primary Source
Direct U.S. supplier	Country	USITC DataWeb
U.S. import quantity	kg / applicable unit	USITC DataWeb
Customs value	USD	USITC DataWeb
Direct supplier share	%	Calculated
Direct-supplier HHI	Index	Calculated
Supplier's own import sources	Country and quantity	National trade statistics
Intermediary upstream shares	%	Calculated
True-source attributed quantity	kg / applicable unit	Calculated
True-source share	%	Calculated
True-source HHI	Index	Calculated
Producer country	Country	USGS
Production quantity/share	Quantity / %	USGS
Production HHI	Index	Calculated where data permit
Processing country	Country	USGS / government sources
Processing concentration	Share / HHI	Calculated where data permit
Dominant-country share	%	Calculated
Net import reliance	% or published bound	USGS
Unit value	Customs value / quantity	Calculated
Export-control event	Date / restriction	Official government sources / USGS
Classification version	Rule/version identifier	TSDA documentation

10. Standard TSDA Output

Measure	2019-2022 Baseline	2023-2025	Change
Direct-supplier HHI	-	-	-
Dominant direct-supplier share	-	-	-
True-source HHI	-	-	-
Dominant true-source share	-	-	-
Production HHI	-	-	-
Dominant producer share	-	-	-
Processing concentration	-	-	-
Net import reliance	-	-	-
Import quantity	-	-	-
Unit value	-	-	-
TSDA Classification			-

11. Principal Methodological References

1. Brainard, J.L., Gulley, A.L., and Nassar, N.T. (2026). "Indirect mineral import reliance and provenance." Mineral Economics. DOI: 10.1007/s13563-026-00629-8. U.S. Geological Survey Publications Warehouse. <https://pubs.usgs.gov/publication/70278951>
2. U.S. Geological Survey (2026). Mineral Commodity Summaries 2026 - Gallium. Includes tariff classifications, 2025 net import reliance, import sources, and world production information. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-gallium.pdf>
3. U.S. Geological Survey (2026). Mineral Commodity Summaries 2026 - Germanium. Includes tariff classifications, import sources, and net import reliance reported as >50%. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026.pdf>
4. U.S. Geological Survey (2024). Quantifying Potential Effects of China's Gallium and Germanium Export Restrictions on the U.S. Economy. Open-File Report 2024-1057. <https://pubs.usgs.gov/publication/ofr20241057>
5. U.S. International Trade Commission. DataWeb: U.S. Trade and Tariff Data. <https://dataweb.usitc.gov/>
6. Statistics Canada. Canadian International Merchandise Trade Web Application. Provides HS10 import data and HS8 export data. <https://www150.statcan.gc.ca/n1/pub/71-607-x/71-607-x2021004-eng.htm>
7. Japan Customs, Ministry of Finance. Trade Statistics of Japan - Code Lists and Data Downloads. Japan uses 9-digit statistical codes consisting of the 6-digit HS code plus a 3-digit domestic code. https://www.customs.go.jp/toukei/sankou/code/code_e.htm
8. Eurostat. International Trade in Goods - Comext Database. Provides detailed trade data by reporting country, partner country, value, quantity, and product classification. <https://ec.europa.eu/eurostat/web/international-trade-in-goods/database>
9. United Nations Statistics Division. HS 2017 Code 811292 classification detail. <https://unstats.un.org/unsd/classifications/Econ/Detail/EN/2089/811292>
10. U.S. Department of Justice and Federal Trade Commission (2023). Merger Guidelines, Guideline 1. HHI above 1,800 is described as highly concentrated in the merger-analysis framework. <https://www.justice.gov/atr/merger-guidelines/applying-merger-guidelines/guideline-1>

Research protocol note: This specification defines the analytical rules for the initial TSDA application. Data-source inventories, data dictionaries, code, and output tables will be maintained separately under the corresponding TSDA research record.